# PROJECT PROPOSAL

## **Title:** Indian Premier League (IPL) Match Data Analysis

### 1. Student Details


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### 2. Problem Statement

In the sports industry, particularly cricket, teams and analysts face challenges in understanding player performance, team strategies, and match-turning moments. Factors like run rates in specific overs, wicket-taking patterns, and extra runs can significantly influence match outcomes.

The aim of this project is to analyze ball-by-ball IPL data to understand batting and bowling patterns, identify key performers under pressure, and find factors that lead to team victories. This analysis will help in providing actionable insights for team strategy and performance evaluation.

**3. Domain**

Sports Analytics / Data Science

**4. Dataset Description**

- **Source:** Uploaded CSV (IPL - DA.csv)
- **Dataset Features:** The dataset contains ball-by-ball information for IPL matches, including:
    - **Match & Delivery IDs:** Unique identifiers for each match and ball.
    - **Innings & Over Details:** Tracking the progression of the game.
    - **Team Information:** Batting and bowling team names (e.g., Kings XI Punjab, Royal Challengers Bangalore).
    - **Player Roles:** Identifies the striker, non-striker, and bowler.
    - **Run Tracking:** Specifically records batsman runs, extra runs, and total runs.
    - **Wicket Data:** Captures if a wicket fell, the dismissal type (e.g., caught, bowled), and players involved.

**5. Objectives**

- To analyze batting trends and strike rates of different players.
- To identify the most effective bowlers based on wicket-taking ability and run containment.
- To study the impact of "extras" (wide, no-ball, etc.) on the final score.
- To visualize match progress through over-by-over run analysis.
- To generate insights into the most common dismissal types in different match phases.

**6. Methodology**

1. **Data Collection:** Use the provided IPL ball-by-ball dataset.
2. **Data Preprocessing:** * Handle missing values in columns like extra_type or fielder.

    - Ensure numeric types for runs and ball counts.
3. **Exploratory Data Analysis (EDA):** * Calculate total runs per match and innings.
    - Identify top-scoring batsmen and leading wicket-takers.
4. **Data Visualization:** * **Bar Charts:** Top run-scorers and wicket-takers.
    - **Pie Charts:** Distribution of dismissal types.
    - **Line Plots:** Run rate progression over 20 overs.
5. **Tools & Technologies:** Python, Pandas, Matplotlib, Seaborn, Jupyter Notebook.

**7. Expected Outcome**

- Comprehensive understanding of player performance metrics.
- Identification of patterns that lead to high-scoring innings.
- Visual representation of team performance across different matches.

**8. Conclusion**

This project will help in understanding the dynamics of T20 cricket using data analysis techniques. The insights gained can be used for performance benchmarking and strategic planning in professional cricket.